PROJECT PROPOSAL & QUOTATION

Digital Certificate Verification & Student Attendance Portal

Independent, Turnkey Web Application with Zero Monthly Cloud Hosting Costs

CLIENT / ORGANIZATION:	HAIFOS Innovative Forum / Partner Organization	TOTAL PROJECT COST:	¹10,000 INR (All-Inclu
PREPARED FOR:	Executive Management & Training Leadership	DELIVERY TIMELINE:	3 to 5 Business Days
RECURRING SERVER COST:	¹0 / Month Forever ($0 Free Tier)	ARCHITECTURE TYPE:	100% Free-Tier (Netlify + Supabase)

1. Executive Summary

This proposal outlines the end-to-end development, testing, deployment, and handover of a dedicated digital web portal engineered specifically for HAIFOS educational initiatives and community training centers.

Traditional certificate management and attendance tracking platforms present significant bottlenecks: they either require expensive monthly software-as-a-service (SaaS) subscriptions, expensive cloud file storage for storing heavy PDF files, or laborious manual email distribution.

Our solution completely re-engineers this model by delivering a custom-built, production-grade portal that achieves 100% Zero Recurring Hosting Costs. Standardized certificates are generated dynamically on-the-fly directly in the browser from database metadata, requiring zero server disk storage. Furthermore, certificate email delivery is seamlessly integrated with a 1-click dispatch directly from your official email client (Gmail / Desktop Mail), eliminating third-party email API fees and delivery failures.

Dynamic Verification

ZERO FILE STORAGE

Certificates render live on demand. Authentic QR code & verification seal. Instant PDF/PNG download.

Daily Attendance

TRANSPARENT REGISTRY

1-click daily batch marking. Cumulative attendance % calculation with certificate eligibility status.

1-Click Direct Email

ZERO API CHARGES

Admin clicks send; pre-filled email template opens in Gmail/mail client. Click send and done.

2. How the System Operates 100% Free Tier ($0 Recurring Cost)

Every layer of the application is designed to stay permanently within high-generosity free tiers without compromising enterprise speed or

1. Frontend Hosting & CDN

Provider: Netlify (Free Tier)

Capacity: 100 GB Monthly Bandwidth • Automated Continuous Git Deployments • Free Global SSL

Cost Advantage: Pure client-side Single Page Application (SPA). Zero backend compute cost. Deployed straight from GitHub.

2. Cloud Database & Sync

Provider: Supabase PostgreSQL (Free Tier)

Capacity: 500 MB Database • 50,000 Monthly Active Users • Instant REST API with Row-Level Security

Cost Advantage: Only structured text rows (name, date, status, cert ID) are stored. 500 MB holds over 300,000+ student records.

3. Certificate Generation Engine

Provider: Client-Side HTML5 Canvas / jsPDF

Capacity: Unlimited Certificate Generations • Zero Server Disk Usage • Zero Cloud Storage Fees

Cost Advantage: The server NEVER stores static PDF or image files! The certificate re-generates dynamically in the user browser.

4. Student Email Dispatch

Provider: Direct 1-Click Mail Client Integration

Capacity: Unlimited Emails • Sends from your own official Gmail / Outlook / Corporate Address

Cost Advantage: No third-party email API subscriptions (e.g. SendGrid/Mailgun) required. Pre-fills message ready for 1-click send.

Financial Comparison Over 3 Years:

EXPENSE CATEGORY	TRADITIONAL LMS / SAAS SYSTEM	OUR PROPOSED SOLUTION
Cloud Server & Storage (AWS/DigitalOcean)	¹1,500 – ¹3,000 / month	**¹0.00 / month (Netlify)**
Transactional Email Subscriptions	¹800 – ¹1,500 / month	**¹0.00 / month (1-Click Direct)**
Database Maintenance & S3 File Storage	¹1,000 / month	**¹0.00 / month (Supabase 500MB)**
3-Year Total Infrastructure Cost	**¹1,18,800+ INR**	**¹0.00 INR (100% Free Forever)**

3. Detailed Functional Modules & Deliverables

Module 1: Public Dynamic Certificate Verification Portal

- 3-Point Security Verification: Requires Course Selection + Certificate ID + (Date of Birth OR Last Name).
- Standardized HAIFOS Certificate Design: Features ornate gold border, Section 8 insignia, and academic serif typography.
- Live Dynamic QR Code: Real-time QR generated via qrcode library linking directly to the public verification URL.
- Zero-Storage Vector Export: High-definition PDF and PNG download buttons generated client-side in seconds.
- Professional Social Sharing: Direct "Add to LinkedIn" credential button and 1-click URL copy feature.

Module 2: Daily Attendance Management & Student Tracking

- Daily Session Logger: Admin selects date and course to automatically load enrolled students.
- 1-Click Batch Marking: Quick "Mark All Present" button alongside individual toggles (Present, Late, Absent, Excused).
- Student Attendance Query: Students search by Enrollment Number, Cert ID, or Email to view their attendance record.
- Visual Percentage Gauge: Instant eligibility badge (>= 75% Eligible for Certificate, < 75% Warning/Defaulter).
- Chronological Session History: Table listing date, topics covered, and instructor verification stamp.

Module 3: Admin Command Center & Security Controls

- Password-Protected Access: Configurable administrator PIN/password to safeguard sensitive student data.
- Student User Management: Add, edit, filter, search, and delete student profiles with unique enrollment numbers.
- Single-Click Certificate Generation: Select student and course to instantly mint an authenticated Certificate ID.
- Dual-Mode Reliability: Built-in local offline demo database + 1-click cloud sync to Supabase PostgreSQL.
- One-Click JSON Backup & Restore: Complete database export and import to safeguard data at all times.

Module 4: 1-Click Direct Email Dispatch (Zero API Cost)

- Zero-Friction Workflow: Clicking "Send Certificate Email" prepares a personalized, branded email template.
- Opens in Your Email Client: Automatically opens Gmail Web or your default desktop mail app with 1 click.
- Pre-Filled Fields: Student email, subject line, congratulations message, and direct verification link are pre-loaded.
- Effortless Review: The admin simply reviews the message on their own screen and clicks "Send" — completely free.

4. Commercial Quotation & Milestone Breakdown

Fixed, transparent pricing with complete end-to-end testing, deployment, and zero hidden costs.

ITEM #	DELIVERABLE DESCRIPTION	TIMELINE	AMOUNT (INR)
1.0	**Portal Frontend & Brand System** Responsive web interface styled to HAIFOS brand, Cinzel academic typography, dual core tabs.	Day 1	¹ 3,500
2.0	**Dynamic Certificate & QR Verification Engine** Zero-storage live generation, 3-point credential validation, high-res PDF & PNG vector export.	Day 2	¹ 2,500
3.0	**Daily Attendance System & 1-Click Emailing** Batch attendance logger, student percentage calculation, 1-click Gmail/mail client template dispatch.	Day 3	¹ 2,000
4.0	**Supabase Cloud Database & RLS Security** PostgreSQL schema setup, multi-device live synchronization, Row-Level Security policies.	Day 4	¹ 1,000
5.0	**Netlify Production Deployment & Handover** Continuous Git build setup, custom domain SSL linking, cross-device testing & admin guide.	Day 5	¹ 1,000
TOTAL FIXED PROJECT COST:			**¹ 10,000 INR**

Payment Terms & Delivery Commitment:

- Fixed Cost Commitment: Total payable is strictly ¹10,000 INR. No extra or surprise charges.
- Recurring Hosting Cost: ¹0.00 / Month Forever (runs on 100% Free Tiers of Netlify and Supal
- Payment Structure: 50% Advance upon project kickoff (¹5,000) • 50% upon final live dep
- Warranty & Bug Support: Includes 14 Days of free post-launch support and live admin training session.
- Complete Ownership: Full Git repository source code and database ownership transferred to your organization.

ACCEPTED & APPROVED BY (CLIENT):

Signature & Official Seal • Date

SERVICE PROVIDER / LEAD DEVELOPER:

Full Stack Solution Lead • Date